

Understanding Your Carbon Footprint

Simple Environmental Science Guide & Fun Activity Worksheet for Students

1. Easy Lesson Recap: What is a Carbon Footprint?

Just like walking on wet mud leaves visible footprints, our daily choices leave an **invisible carbon footprint** in the air! Whenever we burn petrol, use electricity from coal, or buy heavily packaged items, green-house gases like **Carbon Dioxide (CO₂)** go into the atmosphere, making Earth warmer.

The 4 Pillars of Carbon:

-  **Transport:** Cars and buses burning fuel.
-  **Electricity:** Coal plants powering our fans, ACs, and TV.
-  **Food:** Snacks transported from far away vs. local fresh mandi fruit.
-  **Waste:** Plastic and trash dumped in big landfills.

2. Color Zone Checklist

Zone	What It Means	Simple Daily Examples
GREEN (Low Footprint)	Super Eco-Hero! Great choices!	Riding a cycle, taking bucket baths, eating local fruit.
YELLOW (Medium Footprint)	Good effort, but room to improve!	Using carpools, turning off fans when leaving the room.
AMBER (High Footprint)	Needs Action! Lots of waste.	Solo car rides everywhere, 20-min hot shower, single-use plastic.

3. Expanded Student Activity Worksheet

Q1. Fill in the Blanks (Easy Vocabulary Check)

- The main invisible gas that causes our carbon footprint is _____ (CO₂).
- Commuting by _____ (Delhi Metro / Solo Car) creates a much smaller carbon footprint per person.
- Buying fruits from a local _____ cuts down transport emissions.

Q2. Direct vs. Indirect Emissions (True / False or Categorize)

Hint: Direct happens right in front of you; Indirect happens at a distant factory or power plant!

- Exhaust smoke coming out of a petrol scooter on the road: [_____]
- Electricity used to run a room AC coming from a coal power plant: [_____]
- Garbage decomposing in a landfill emitting gases: [_____]

Q3. Match the Eco-Hacks (Jugaad Solutions)

Daily Problem:

- Running shower uses 50 liters of hot water.
- Single-use plastic shopping bags pile up.
- Broken electronics creating e-waste.

Green Jugaad Hack:

- Repurpose old sarees/cloth into shopping bags.
- Take 5-minute bucket baths.
- Repair items locally instead of buying new ones.

Answers: 1 → _____, 2 → _____, 3 → _____

Q4. Aarav's Morning Fix!

Aarav from Jaipur uses an electric heater for 30 minutes, eats chips from a plastic pouch, and gets dropped at school alone in a big SUV car. List **3 simple changes** Aarav can make to jump from the Amber Zone to the Green Zone:

Q5. Home Energy Detective Tracker (3-Day Log)

Check your home electricity meter with a parent morning & evening for 3 days!

Day	Morning Reading	Evening Reading	Top 2 Power-Hungry Appliances Used Today
Day 1			
Day 2			
Day 3			

Q6. Food Miles Challenge

Why does eating fresh local Nagpur oranges produce a smaller carbon footprint than eating imported or plastic-packaged snack foods from far away?

Q7. My Personal Green Pledge

Write down two daily eco-friendly promises you will follow starting today:

1. I pledge to _____
2. I pledge to _____